MITHIL K GOWDA

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Professional Summary

Cybersecurity graduate specializing in Security Operations (SOC), threat detection, security monitoring, incident response, and secure software engineering. Project Intern at ISRO–LEOS with experience developing secure backend applications using Python, Flask, REST APIs, and MySQL. First author of internationally recognized cybersecurity research on AI-driven cyber defense, threat intelligence, and autonomous security architectures. Seeking graduate opportunities in SOC, Blue Team Operations, Threat Intelligence, and Security Engineering.

Education

M.S. Ramaiah University of Applied Sciences, Bengaluru 2022–2026

B.Tech. in Information Science and Engineering

CGPA: **8.80 / 10**

Technical Skills

Security Operations: Security Operations Center (SOC), Threat Detection, Security Monitoring, Incident Response, Threat Intelligence, Threat Hunting, Log Analysis, IOC Analysis, Vulnerability Assessment, Threat Modeling

SIEM & Detection Engineering: Splunk, Microsoft Sentinel, Wazuh, Windows Event Logs, Sysmon, Sigma Rules, YARA, Security Event Correlation

Security Tools: Kali Linux, Wireshark, Nmap, Burp Suite, OWASP ZAP, Nessus, Metasploit, TCPDump

Operating Systems: Windows, Linux, Kali Linux

Networking: TCP/IP, DNS, HTTP/HTTPS, SSH, VPN, DHCP, OSI Model, Network Traffic Analysis

Programming & Scripting: Python, Java, JavaScript, SQL, HTML, CSS, Bash

Frameworks & Standards: MITRE ATT&CK, OWASP Top 10, NIST Cybersecurity Framework, Security Architecture, REST APIs

Databases: MySQL, MongoDB

Cloud: Google Cloud Platform (GCP)

Professional Experience

Indian Space Research Organisation (ISRO) – Laboratory for Electro Optics Systems (LEOS), Bengaluru
2026

Project Intern – Secure Software Development & Data Analytics

- Designed and developed secure backend applications using Python, Flask, REST APIs, and MySQL to support analytics-driven engineering workflows while implementing secure coding practices.
- Implemented authentication mechanisms, secure database integration, role based access workflows, and application security controls to strengthen confidentiality, integrity, and controlled system access.
- Built secure data processing and visualization pipelines supporting operational analytics, structured data management, and engineering decision-making.
- Developed scalable and secure backend application modules emphasizing secure software architecture, maintainability, operational reliability, and protected backend communication.

Cybersecurity Projects & Research

A Multi-Agent AI Cyber Defense Framework Using HoneyBee Method

- First Author and Corresponding Author of an AI-driven cybersecurity framework accepted for presentation at ICIICE 2026 (Dubai, UAE) and accepted as a book chapter in the edited volume *Metaheuristic Optimization for Social Good.*
- Designed a multi-agent cyber defense architecture supporting enterprise threat detection, security monitoring, adaptive deception, threat intelligence collection, autonomous incident response, and continuous learning.

- Engineered the Detect–Mislead–Neutralize–Learn security lifecycle integrating XGBoost, CNN-LSTM, Deep Q-Network (DQN), adaptive honeypots, distributed threat intelligence, and defensive AI.
- Evaluated the framework using CIC-IDS2017 datasets and simulated enterprise attack scenarios, achieving 98.24% detection accuracy, 98.51% precision, 98.10% recall, 0.992 ROC-AUC, and 1.85% false positive rate.

AI-Governed Carbon Credit Exchange with Digital Carbon Passport

- Abstract accepted at the 4th International Conference on Advancing Sustainable Futures (ICASF 2027), Abu Dhabi University, UAE.
- Designed a secure distributed application integrating AES-256 encryption, SHA-256 hashing, authentication mechanisms, fraud detection, trust scoring, and secure transaction validation for digital asset protection.
- Developed the Secure Trade Authorization and Verification Pipeline (STAVP) utilizing AES-256 encryption, SHA-256 hashing, authentication mechanisms, and secure transaction verification.

C^3T-STAVP Cryptographic Framework (Ongoing Research)

- Designing a hybrid cybersecurity framework integrating Character Chaffing & Transposition Technology (C^3T), Secure Transaction Authorization & Verification Protocol (STAVP), and adaptive trust validation.
- Researching Zero-Knowledge Proofs (ZKP), post-quantum cryptography, semantic obfuscation, secure transaction validation, privacy-preserving architectures, and resilient defensive security mechanisms.

Professional Certifications

- IBM IT Fundamentals for Cybersecurity Specialization – IBM
- Ethical Hacker – Cisco Networking Academy
- Introduction to Cyber Security – IGNOU (SWAYAM)
- Systems and Usable Security (Elite) – NPTEL
- Google Cloud Computing – NPTEL
- Introduction to Internet of Things (Silver Medal) – NPTEL
- Business Intelligence and Analytics – NPTEL
- IBM Front-End Developer Specialization – IBM

Achievements

- Selected as Project Intern at the Indian Space Research Organisation (ISRO – LEOS).
- First Author of internationally recognized cybersecurity research focused on AI-driven cyber defense, autonomous threat detection, and intelligent security architectures.
- Awarded the Reliance Foundation Undergraduate Scholarship (2026).
- Recipient of NPTEL Silver Medal in Introduction to Internet of Things.